

PARTH PANDYA

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EXPERIENCE

Stack Summation

Jul. 2022 – Jun. 2024

- Web Application Developer* | Python, Django, FastAPI, Vue.js, PostgreSQL, REST API, Git, CI/CD India
- Architected scalable backend services using **Python (Django)** and **FastAPI**, delivering **production-grade REST APIs** for concurrent client workflows and automated reporting pipelines.
 - Optimized complex **SQL queries** (joins, aggregations, indexing, stored procedures), reducing latency and improving application performance by **~30%**.
 - Integrated **Google Maps, Drive, and PayPal APIs** for geospatial, cloud storage, and payment features serving **500+ users**; engineered webhook callbacks, idempotency handling, and failure recovery.
 - Built **Vue.js** frontend with dynamic charts for resort weather analytics; resolved critical **PHP** and frontend defects in U.S.-based production system via **CI/CD pipelines** and centralized logging.
 - Implemented **role-based access control (RBAC)** and session management across multi-tenant workflows, improving system security posture and enforcing least-privilege principles in production.
 - Containerized application services using **Docker** and configured **GitHub Actions** workflows for automated build, test, and deployment pipelines, reducing manual release effort and deployment errors.

Mindtrunk

Jul. 2021 – Jun. 2022

- Backend Developer* | Python, FastAPI, SQLAlchemy, Vue.js, AWS S3, Qt5, PostgreSQL India
- Developed real-time **image analysis pipelines** on **Raspberry Pi**, improving edge-device processing speed by **30%**; designed **multi-engine database systems** via **SQLAlchemy ORM** across SQLite, MySQL, and PostgreSQL.
 - Built **FastAPI** services cutting response times by **~50%**; integrated **AWS S3** (load times reduced 40%); engineered **Qt5 computer vision** and **Vue.js** frontend boosting engagement by **30%**; validated **4x** peak traffic scale.
 - Authored **unit and integration test suites** using **pytest**, achieving **85%+ code coverage** and reducing post-release defect rate through systematic regression and boundary testing.
 - Designed and documented **RESTful API contracts** using OpenAPI/Swagger, enabling seamless frontend-backend integration and reducing cross-team miscommunication during iterative development cycles.

LEADERSHIP AND ACADEMIC CONTRIBUTIONS

Tennessee State University, Dept. of Electrical and Computer Engineering

Aug. 2024 – Apr. 2026

- Teaching Assistant and Research Assistant* USA
- Conducted research across **AI, sensor fusion, and data-driven modeling**; designed and benchmarked **machine learning and statistical models** in **Python and MATLAB** with rigorous validation, profiling, and reproducibility standards.
 - Engineered **end-to-end data preprocessing, feature engineering, and visualization pipelines** with automated quality checks ensuring statistically validated model inputs and outputs.
 - Developed **memory-safe C programs** with low-level tuning and **TCP/IP and UDP socket** modules for distributed and networked system architectures; executed lab instrumentation, calibration, and data acquisition.
 - Instructed undergraduate courses in **Probability and Statistics, Python Programming, and MATLAB**; delivered feedback and tutoring on **OOP, numerical methods, and algorithm design** across diverse learning profiles.

TECHNICAL SKILLS

Languages: Python, Bash, PowerShell, Java, C, C++, SQL **CI/CD and DevOps:** Jenkins, GitLab CI, GitHub Actions, Git, Linux **Cloud and Infrastructure:** AWS (EC2, S3, Lambda), Azure, GCP, Docker, Kubernetes, Terraform, Ansible **Observability:** Prometheus, Grafana, Centralized Logging

Backend and APIs: REST API, Microservices, FastAPI, Django, Multithreading, OOP, Design Patterns **Databases:** PostgreSQL, MySQL, SQLite, Query Optimization

Testing and Quality: Unit and Integration Testing, Debugging, Root Cause Analysis, Technical Documentation

EDUCATION

Tennessee State University

Aug. 2024 – May 2026

Master of Science, Computer and Information Systems Engineering | GPA: 3.36/4.0

Nashville, TN

Gujarat Technological University

Aug. 2017 – May 2021

Bachelor of Engineering, Information Technology | GPA: 3.59/4.0

Gujarat, India

PROJECTS

Car Rental System | Python, Django, MySQL, REST API, OAuth, HTML, CSS, JavaScript, AWS

Spring 2024

- Full-stack platform with **OAuth** (Gmail/Facebook), real-time booking, **automated email and SMS notifications**, payment gateway with webhook and failure recovery; deployed on **AWS/Azure** with SSL and role-based Admin Dashboard.

Pattern Recognition: Hopfield Memory Networks | Python, MATLAB, NumPy, Matplotlib

Spring 2025

- Hopfield Neural Network from scratch** using **Hebbian learning**; evaluated energy minimization, spurious attractors, and retrieval accuracy across noise configurations; visualized recall dynamics via **Matplotlib**.

CERTIFICATIONS AND COURSEWORK

Certifications: Certified in C, C-DAC GIST (Apr. 2018); Certified in C++, C-DAC GIST (Apr. 2019)

Relevant Coursework: System Modeling, Systems Engineering, Software Systems Design, Artificial Intelligence, Digital Image Processing, Computer Architecture